

VED NARVEKAR

BACKEND & AI ENGINEER

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SUMMARY

Backend & AI Engineer with hands-on experience building production-grade systems — high-throughput API gateways handling **5,000+ RPS**, autonomous LLM-powered CLI agents with multi-provider support, and ML-backed web applications with measurable performance outcomes. Proficient in Node.js, TypeScript, Python, system design, and applied AI/ML. Seeking backend or AI engineering internships to ship reliable, impactful production features.

EDUCATION

B.E. Artificial Intelligence & Machine Learning — Mumbai University

Expected Apr 2028

Universal College of Engineering, Mumbai | Discipline & Anti-Ragging Committee Head

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, C++
Backend	Node.js, Fastify, Express.js, REST APIs, System Design, Authentication, RBAC
AI / ML	PyTorch, NumPy, Pandas, Scikit-learn, LangChain, RAG Pipelines, LLM APIs
Databases	PostgreSQL, MongoDB, Redis, Drizzle ORM, Database Design
DevOps	Linux, Git, GitHub Actions, Prometheus, Docker, AWS, Postman
Frontend	React, Next.js, HTML, CSS, MERN Stack

EXPERIENCE

Backend Engineer Intern — SR India

Jan 2026 – Feb 2026 · Remote

- Engineered **REST APIs** for a hotel booking platform (Node.js, Express, PostgreSQL) covering **4 core modules** — bookings, room availability, user management, and admin controls — with **JWT auth and RBAC**.
- Improved API performance and reliability through targeted **query optimization**, input validation, and **Redis caching**; eliminated redundant DB calls across high-frequency booking endpoints.
- Integrated all endpoints into a React dashboard and maintained **Postman collections** documenting request/response contracts, enabling zero back-and-forth handoffs with the frontend team.

PROJECTS

resolv — AI-Powered GitHub Issue Resolver CLI

github.com/vednarvekar/resolv-v2

TypeScript · Node.js · LLM APIs · RAG Pipelines · ts-morph

- Autonomous CLI agent that fetches GitHub issues, fingerprints repo coding patterns via **RAG** (ts-morph DNA extraction + NIM embeddings), and generates **style-matched patches** via an LLM tool-calling loop in **under 60 seconds**.
- Unified **7 LLM providers** (Anthropic, OpenAI, Gemini, Grok, OpenRouter, NVIDIA NIM, Ollama) behind a single OpenAI-compatible interface — provider switch requires one config change.
- Shipped web search (Brave + DuckDuckGo fallback), codebase grep/write tools, persistent chat sessions, and repo-context compression cutting **LLM token usage**.

API Gateway — High-Performance Reverse Proxy

github.com/vednarvekar/api-gateway

Node.js · Fastify · TypeScript · Redis · PostgreSQL · Prometheus

- Architected a production-grade API gateway from scratch reaching **~4,000 RPS at 250 concurrent users**, outperforming naive Express proxies by **4×**; load-tested with autocannon.
- Implemented **Redis sliding-window rate limiting**, circuit breaking (opossum), response caching, JWT/RBAC, Prometheus metrics, and Drizzle ORM migrations — reduced downstream service load by **~40%** under burst traffic.

Pixel — AI Media Detection

github.com/vednarvekar/pixel-v2

React · TypeScript · FastAPI · PyTorch · PostgreSQL · Supabase

- Built a full-stack image and video scanner that fuses **ResNet-18 inference**, EXIF/XMP/IPTC and C2PA metadata heuristics, and **Google Vision Web Detection** into a weighted AI-probability score (default **85% model / 10% metadata / 5% web**), verdict, reasoning, and signal breakdown.
- Orchestrated React/Vite, Express/TypeScript, and a FastAPI/PyTorch model service with PostgreSQL scan history and Supabase email/Google OAuth; supports media up to **10 MB images** and **100 MB videos** with 30-second video analysis.

CERTIFICATIONS

- Complete Data Science, Machine Learning, DL & NLP Bootcamp
- Node.js, Express, MongoDB & More: The Complete Bootcamp
- The Ultimate React Course 2025: React, Next.js, Redux & More

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